

The background of the slide is an abstract composition of flowing, organic shapes in shades of blue, teal, and green. These shapes overlap and curve, creating a sense of movement and depth. The colors are vibrant and saturated, with some areas appearing lighter and others darker, suggesting a light source or a gradient. The overall effect is a modern, artistic backdrop for the text.

Section 4

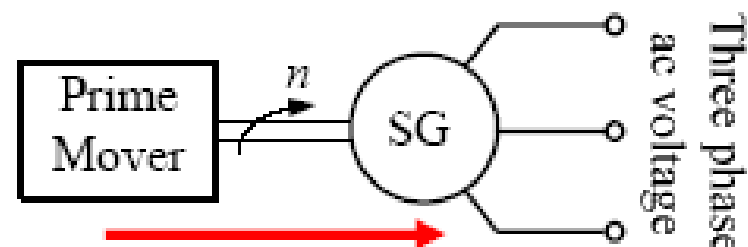
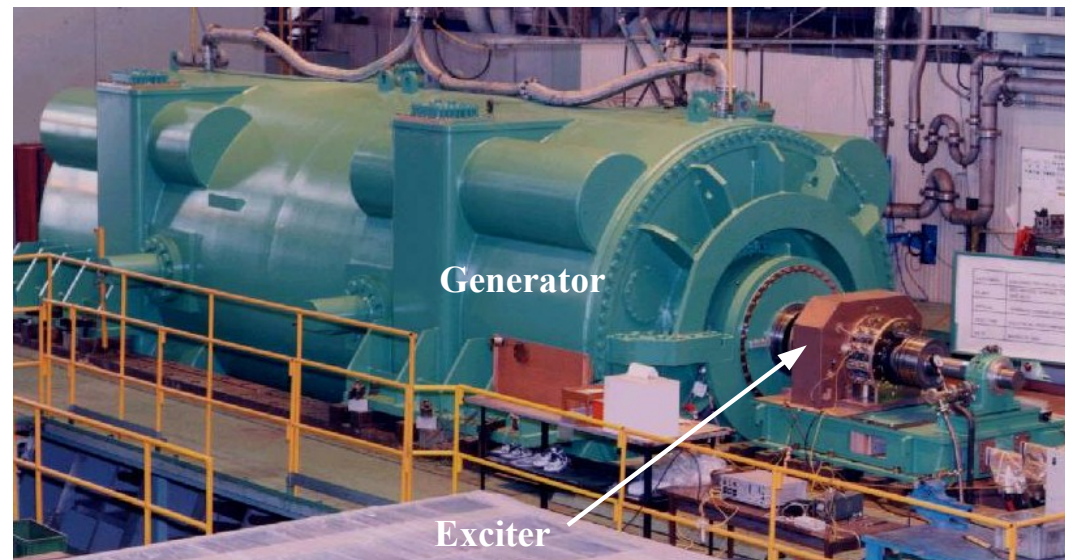
Synchronous Generator

1. Introduction Synchronous Generator

- A **synchronous generator**, also called an **alternator**, is a machine that converts **mechanical power into AC electrical power**.
- It generates electricity by rotating a magnetic field within stationary armature windings.
- Synchronous generators are the **main source** of electrical power in power plants (thermal, hydro, nuclear), diesel generator sets, renewable systems (e.g., wind turbines with synchronous alternators).

Applications

- Power stations (main use)
- Industrial generators
- Diesel generator backup systems
- Marine power systems
- Wind turbine synchronous alternators



2. Synchronous Motor vs. Synchronous Generator

Items	Synchronous Motor	Synchronous Generator
Function	Converts electrical energy → mechanical energy	Converts mechanical energy → electrical energy
Output	Mechanical torque	3-phase AC power
Working Principle	Rotor follows rotating magnetic field of stator	EMF is induced in stator conductors when rotor field cuts them
Rotor Voltage	DC excitation provides magnetic field for torque production	DC excitation provides main field to induce EMF in stator
Starting Method	Not self-starting; needs damper windings or external starter	Self-starting by prime mover

3. Construction

A synchronous generator mainly has four parts:

- **Stator:**

- Contains three-phase armature windings.
- Slots house the copper windings arranged 120° electrically apart.

- **Rotor:**

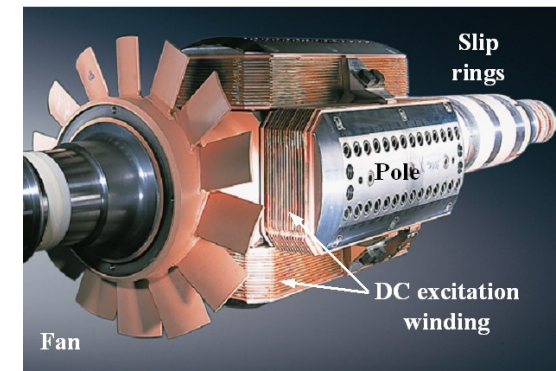
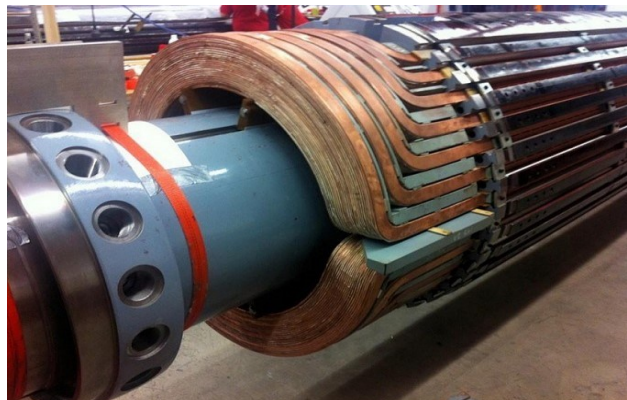
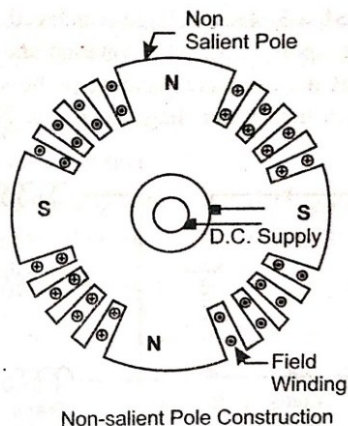
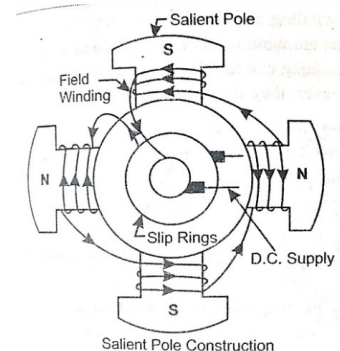
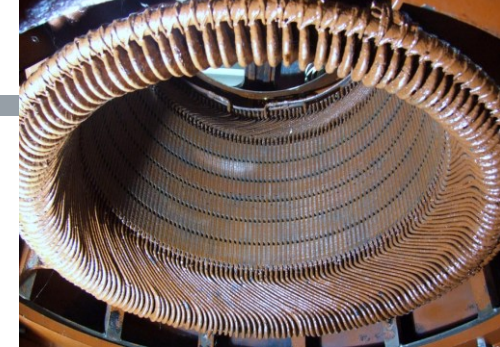
Two types of rotor constructions are common:

- **Salient Pole Rotor**

- Many projecting poles (4 to 40 poles).
- Used in low-speed machines

- **Non-salient (Cylindrical) Rotor**

- Smooth, solid steel cylinder with 2–4 poles.
- Used in high-speed machines



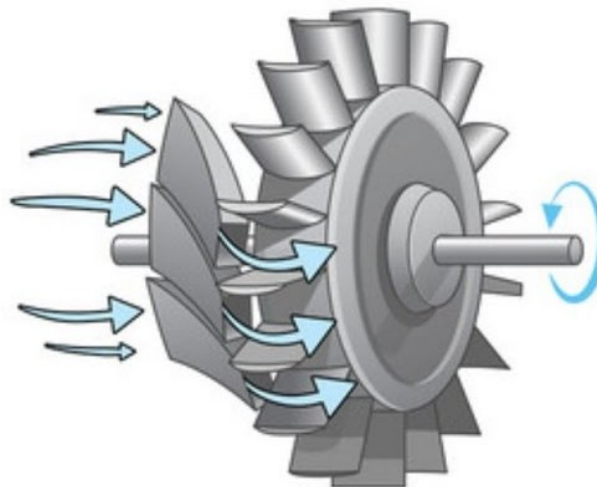
3. Construction

A synchronous generator mainly has four parts:

- **Excitation System:** Supplies DC current to the rotor field winding using slip rings and brushes.



- **Prime Mover:** Provides mechanical input such as turbine (steam, gas, hydro).



4. Working Principle



4. Working Principle

A synchronous generator (alternator) converts **mechanical energy** into **electrical energy** using the principle of **electromagnetic induction**.

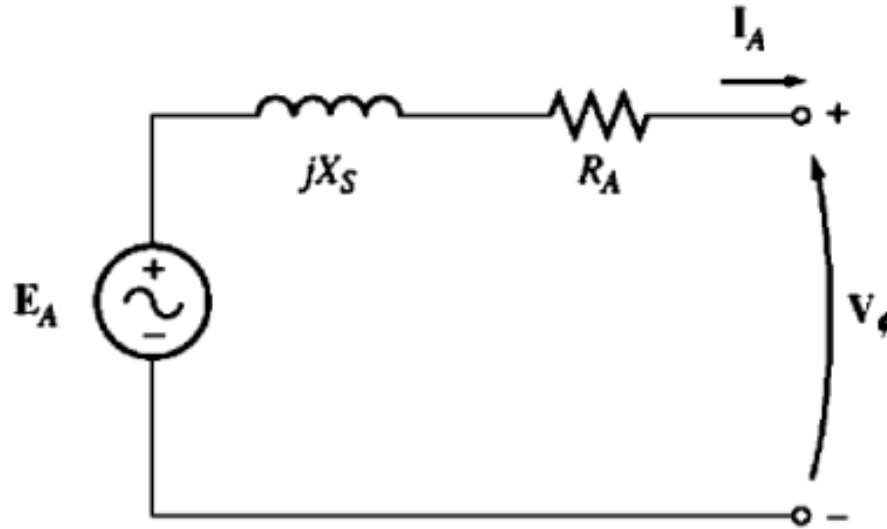
- **Rotor is energized with DC**
 - A small DC generator or rectifier feeds **DC current** to rotor windings.
 - This creates a strong **magnetic field** around the rotor.
- **Prime mover rotates the rotor**
 - The rotor starts spinning at **synchronous speed**:

$$n_{sync} = \frac{120 f_e}{P}$$

- **Rotating magnetic field cuts stator windings**
 - The magnetic poles rotate.
 - They continuously pass across the stator windings.
- **EMF is induced in the stator**
 - The magnetic poles rotate.
 - They continuously pass across the stator windings.
- **Three-phase AC is generated**
 - Each winding gets a sinusoidal voltage.
 - All three voltages are equal in magnitude and 120° phase shifted.

5. The induced voltage in a 3-phase set of coils:

In three coils, each of N turns, placed around the rotor magnetic field, the induced in each coil will have the same magnitude and phases differing by 120° :



$$E_A(t) = N\phi\omega_m \cos(\omega_m t)$$

$$V_\phi = E_A - jX_S I_A - R_A I_A$$

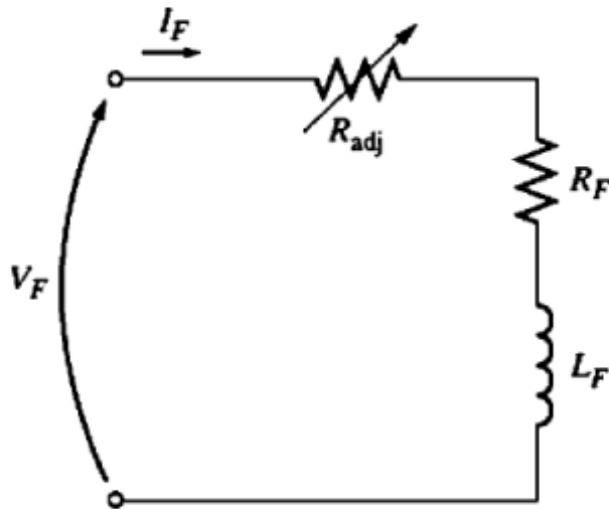
$$E_B(t) = N\phi\omega_m \cos(\omega_m t - 120^\circ)$$

$$E_C(t) = N\phi\omega_m \cos(\omega_m t - 240^\circ)$$

Peak voltage: $E_{\max} = N\phi\omega_m$

RMS:
$$E_{rms} = \frac{E_{\max}}{\sqrt{2}} = \frac{N\phi\omega_m}{\sqrt{2}} = K\phi\omega_m$$

6. The Equivalent Circuit of a Synchronous Generator



$$V_{\phi} = E_A - jX_S I_A - R_A I_A$$

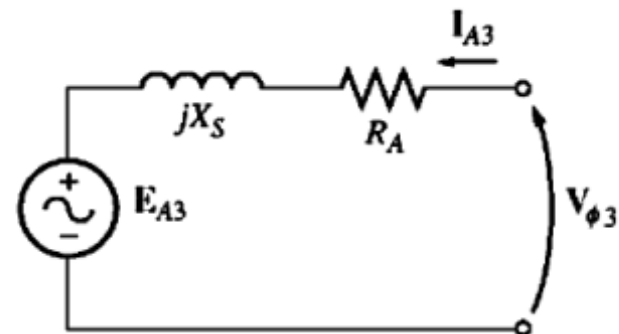
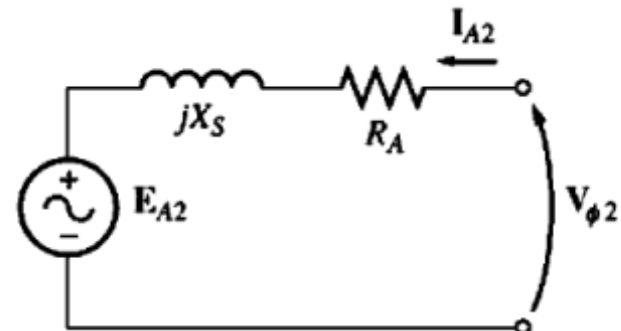
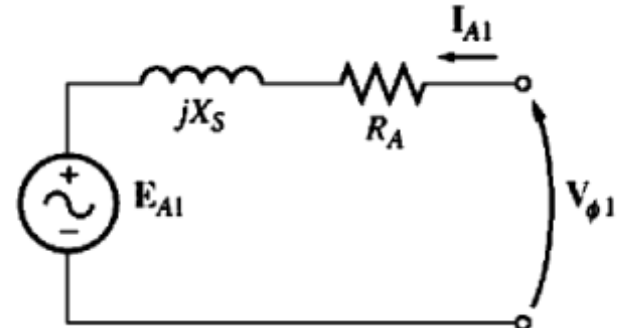
V_{ϕ} : Per-phase applied voltage

E_A : Internal generated EMF

I_A : Armature current

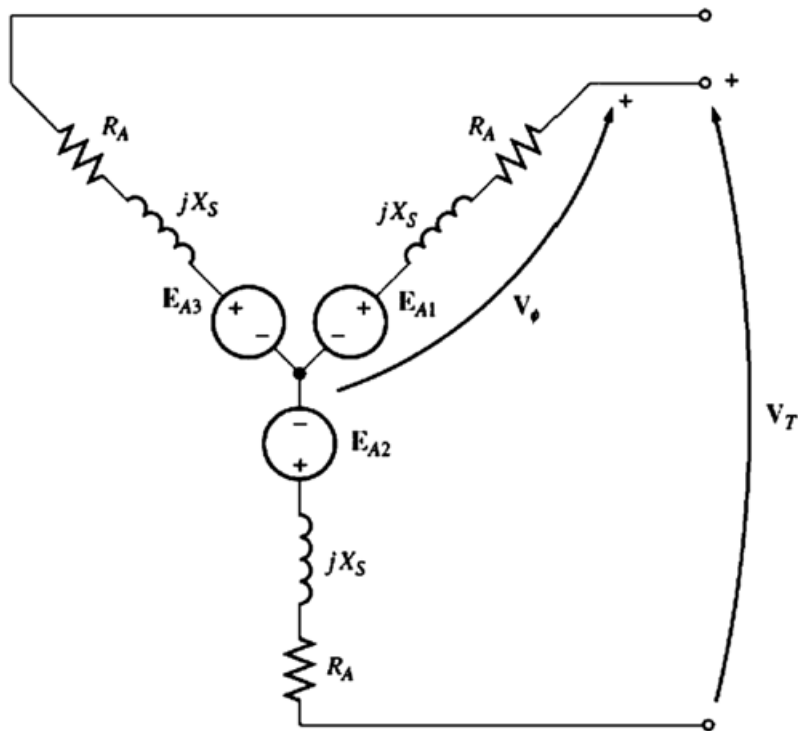
R_A : Armature resistance

X_S : Synchronous reactance



7. Y/ Δ -connection

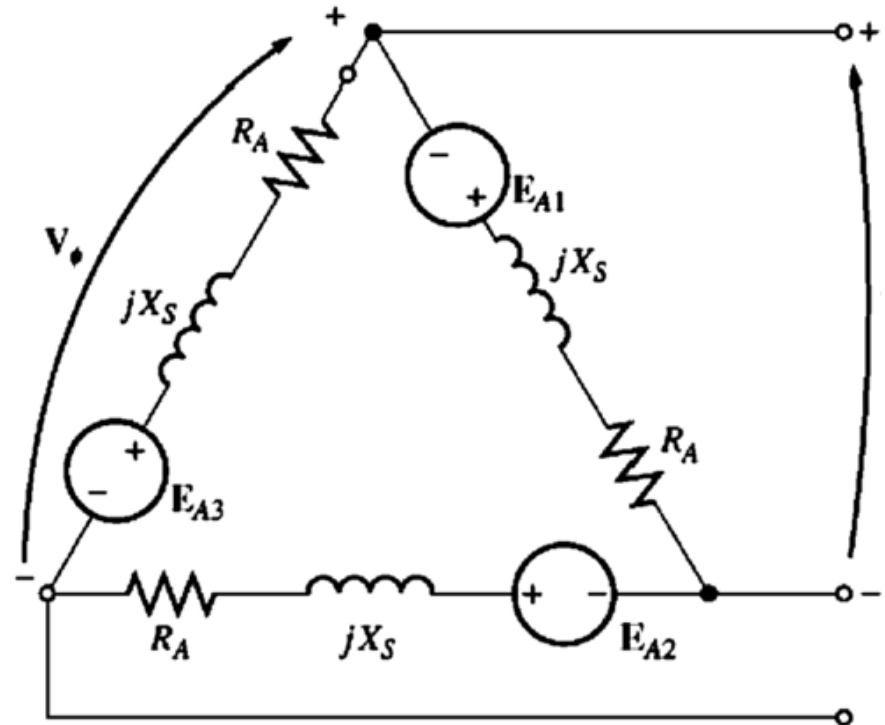
Y-connected



Applications:

- High-Voltage Power Generation
- Long-Distance Power Transmission
- Neutral Availability

Δ -connected

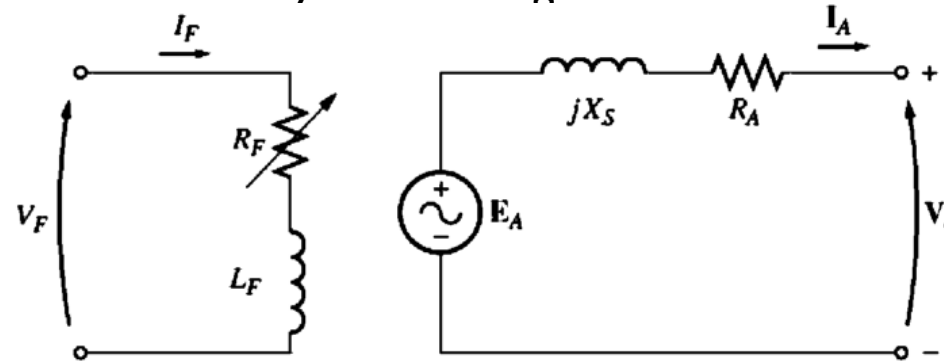


Applications:

- For low-voltage, high-current operation
- Systems Requiring No Neutral
- Parallel Operation of Generators

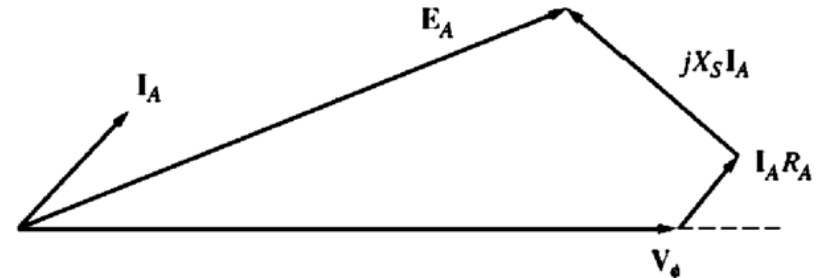
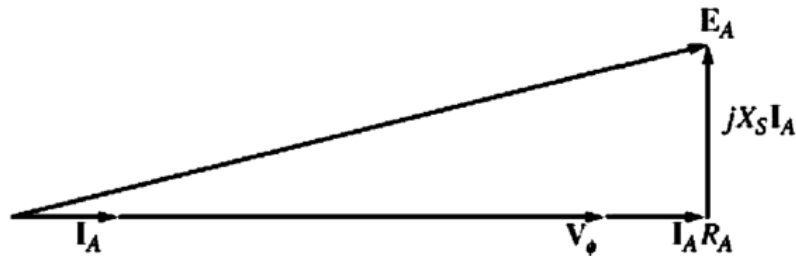
8. The Phasor Diagram

The per-phase equivalent circuit of a synchronous generator:



$$E_A = V_\phi + jX_S I_A + R_A I_A$$

- The phasor diagram at unity power factor
- The phasor diagram at leading power factor

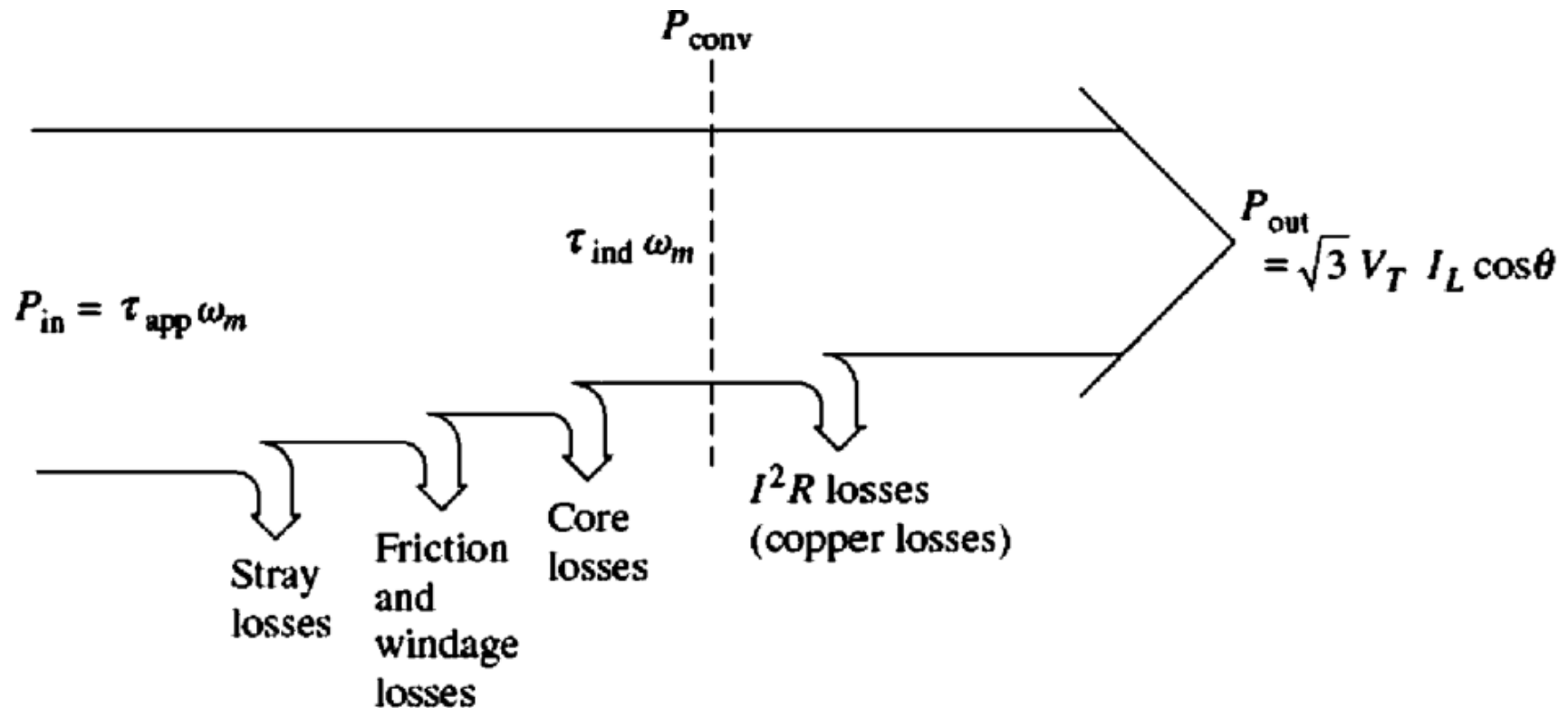


- The phasor diagram at lagging power factor.



9. Power and Torque in Synchronous Generator

- The source of mechanical power, the prime mover, may be a diesel engine, a steam turbine, a water turbine, or any similar device.
- Not all the mechanical power going into a synchronous generator becomes electrical power out of the machine.
- The difference between input power and output power represents the losses of the machine. A power-flow diagram for a synchronous generator is:



9. Power and Torque in Synchronous Generator

- The input mechanical power is the shaft power in the generator:

$$P_{in} = \tau_{app} \omega_m$$

- The power converted from mechanical to electrical form internally is given by:

$$P_{conv} = \tau_{ind} \omega_m$$

$$P_{conv} = 3E_A I_A \cos \gamma$$

- Output power:

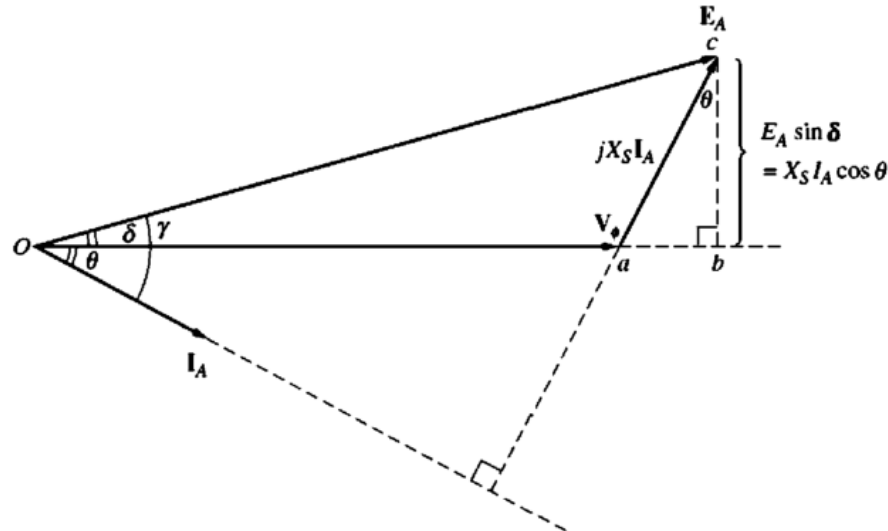
$$P_{out} = \sqrt{3} V_T I_L \cos \theta$$

$$P_{out} = 3V_\phi I_A \cos \theta$$

- In real synchronous machines of any size, the armature resistance $R_a \ll X_s$ and, therefore, the armature resistance can be ignored. Thus, a simplified phasor diagram indicates that:

$$X_s I_A \cos \theta = E_A \sin \delta$$

$$I_A \cos \theta = \frac{E_A \sin \delta}{X_s}$$



9. Power and Torque in Synchronous Generator

- Then the real output power of the synchronous generator can be approximated as:

$$P_{out} = \frac{3V_{\phi} E_A \sin \delta}{X_s}$$

- The maximum power can be supplied by the generator when $\delta = 90^\circ$:

$$P_{max} = \frac{3V_{\phi} E_A}{X_s}$$

- Voltage regulation:

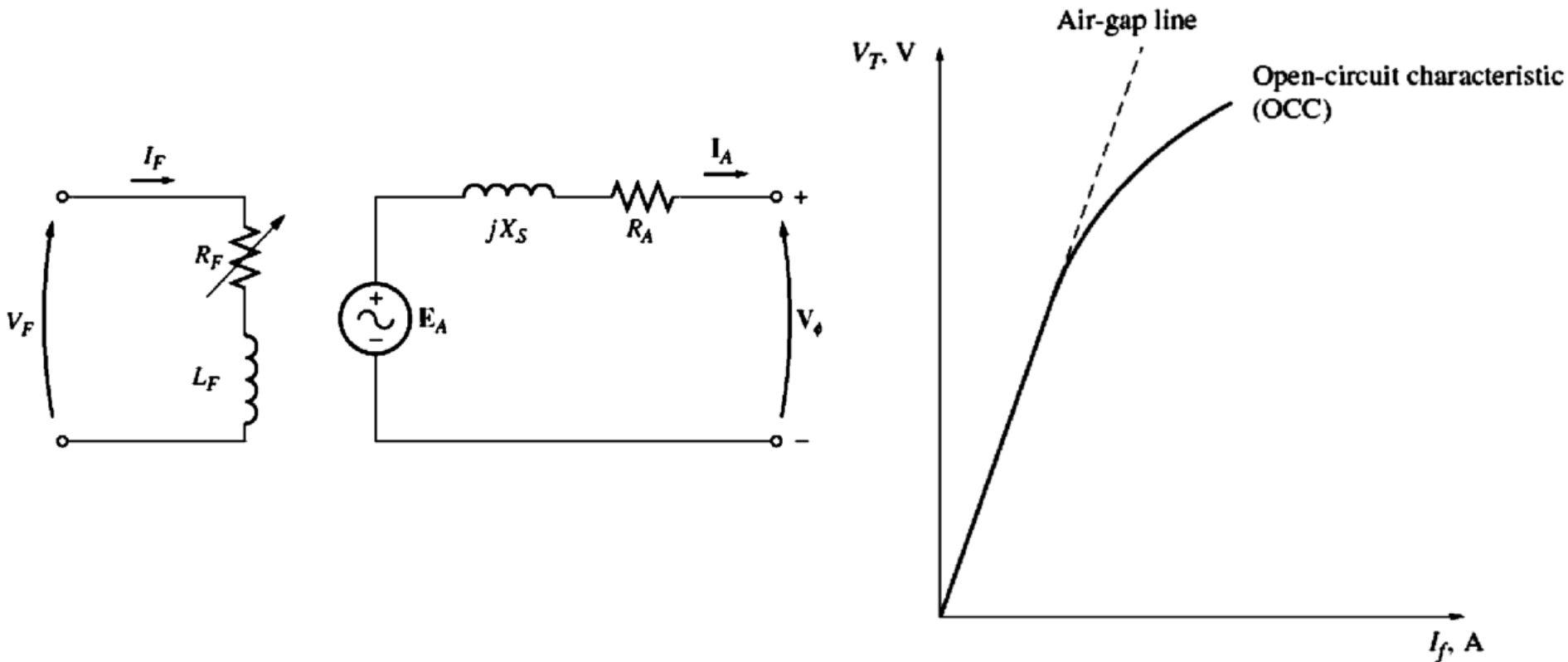
$$\%VR = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

10. Measuring Synchronous Generator Parameters

- **Open-circuit test:**

The open-circuit test, or the no-load test, is performed by:

- Generator is rotated at the rated speed.
- No load is connected at the terminals.
- Field current is increased from 0 to maximum.
- Record values of the terminal voltage and field current value.

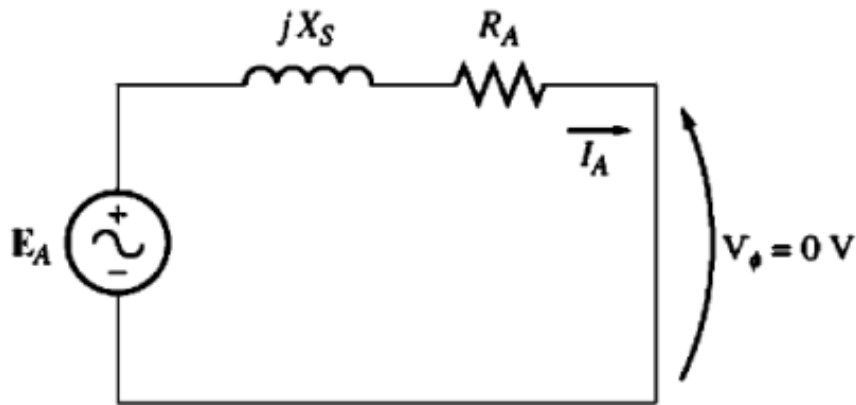


10. Measuring Synchronous Generator Parameters

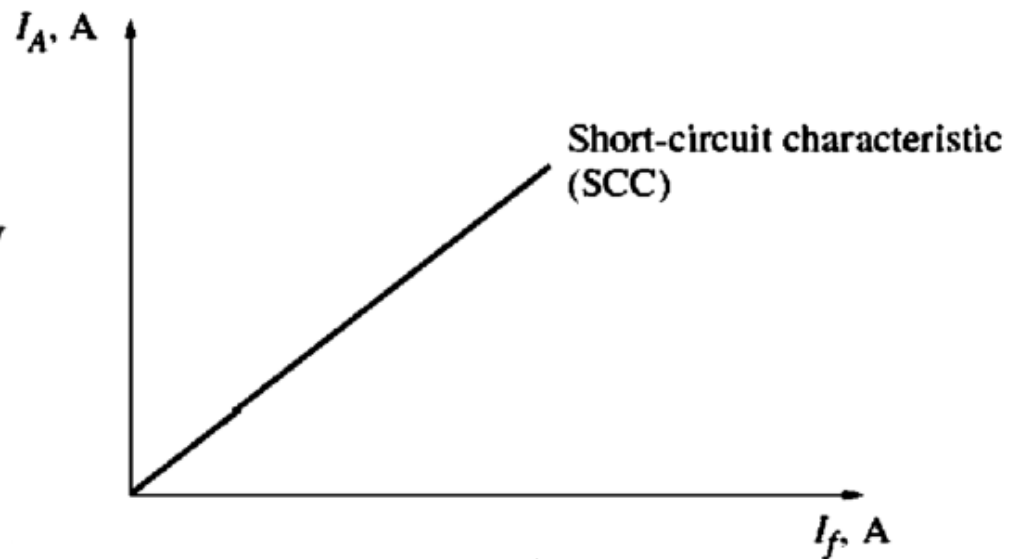
- **Short-circuit test:**

The short-circuit test provides information about the current capabilities of a synchronous generator. It is performed by

- Generator is rotated at rated speed.
- Adjust field current to 0.
- Short circuit the terminals.
- Measure armature current or line current as the field current is increased.



$$I_A = \frac{E_A}{R_A + jX_S}$$



- The resistance can be approximated by applying a **dc voltage** to the windings.
- The use of dc voltage means that the **reactance** of the windings will be **zero** during the measurement process.

Example 1:

A 200-kVA, 480-V, 50-Hz, Y-connected synchronous generator with a rated field current of 5 A was tested, and the following data were taken:

- $V_{T,OC}$ at the rated I_F was measured to be 540 V.
- $I_{L,SC}$ at the rated I_F was found to be 300 A.
- When a dc voltage of 10 V was applied to two of the terminals, a current of 25 A was measured.

Find the values of the **armature resistance** and the approximate **synchronous reactance** in ohms that would be used in the generator model at the rated conditions.

Example 2:

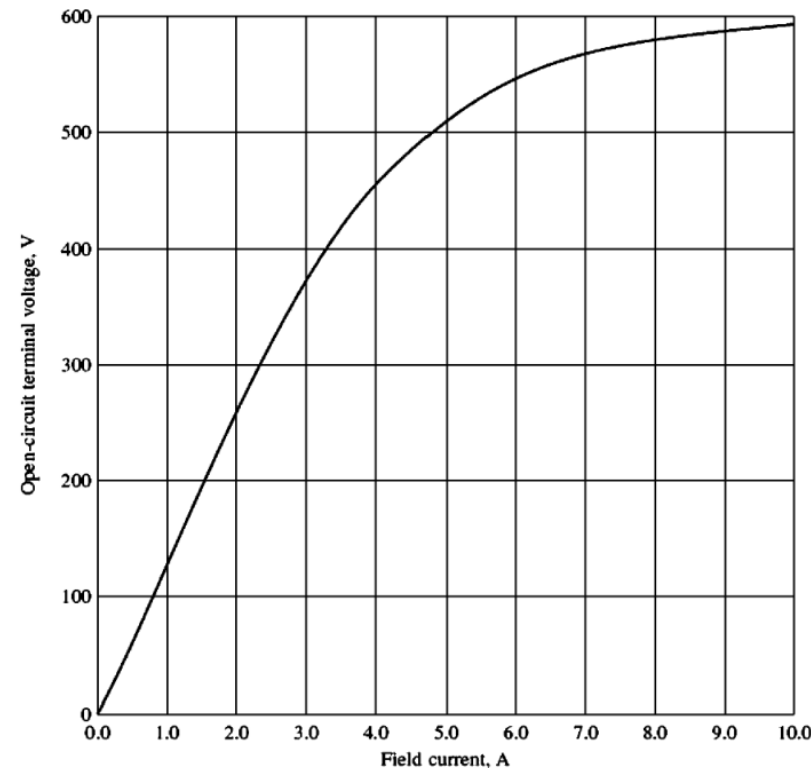
A 480-V, 50-Hz, Y-connected, six-pole synchronous generator has a per-phase synchronous reactance of $1.0\ \Omega$. Its full-load armature current is 60 A at 0.8 PF lagging. This generator has friction and windage losses of 1.5 kW and core losses of 1.0 kW at 50 Hz at full load. Since the armature resistance is being ignored, assume that the I^2R losses are negligible.

1. What is the speed of rotation of this generator?
2. What is the efficiency of this generator when it is operating at the rated current and 0.8 PF lagging?
3. How much shaft torque must be applied by the prime mover at full load?
4. What is the voltage regulation of this generator at 0.8 PF lagging?

Example 3:

A 480-V, 60-Hz, Δ -connected, four-pole synchronous generator has the OCC shown in Figure. This generator has a synchronous reactance of $0.1\ \Omega$ and an armature resistance of $0.015\ \Omega$. At full load, the machine supplies 1200 A at 0.8 PF lagging. Under full-load conditions, the friction and windage losses are 40 kW, and the core losses are 30 kW. Ignore any field circuit losses.

1. What is the speed of rotation of this generator?
2. How much field current must be supplied to the generator to make the terminal voltage 480V at no load?
3. If the generator is now connected to a load and the load draws 1200 A at 0.8 PF **lagging**, how much field current will be required to keep the terminal voltage equal to 480V?
4. How much power is the generator now supplying? How much power is supplied to the generator by the prime mover? What is this machine's overall efficiency?
5. Suppose that the generator is connected to a load drawing 1200 A at 0.8 PF **leading**. How much field current would be required to keep V_T at 480V?



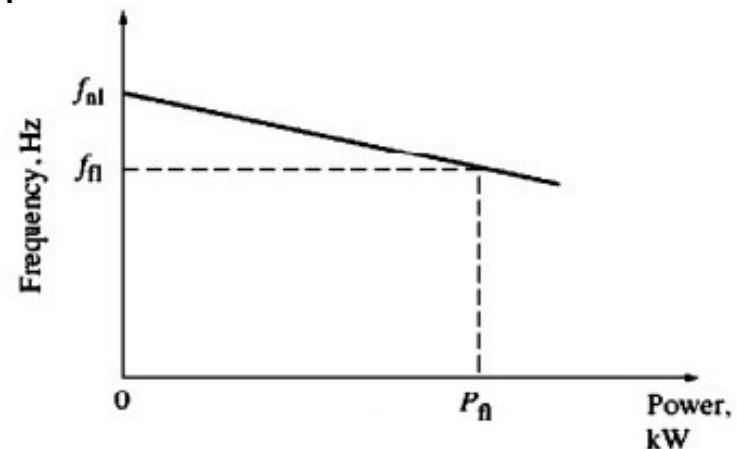
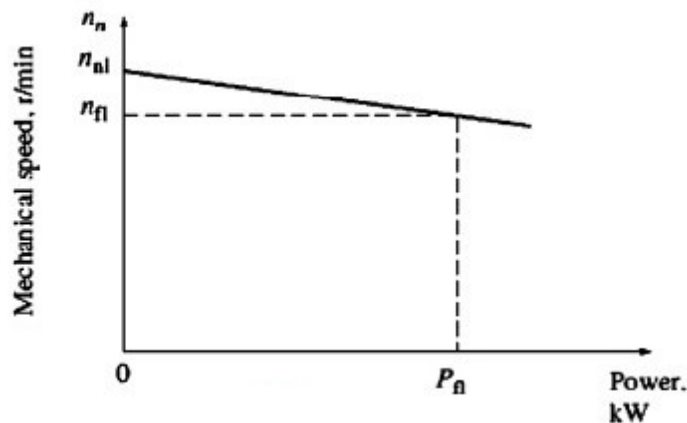
11. Generator P-f Curve

All generators are driven by a prime mover, such as a steam, gas, water, wind turbines, diesel engines, etc.

- Regardless the power source, most of prime movers tend to slow down with increasing the load.
- The speed drop (SD) of a prime mover is defined as:

$$\%SD = \frac{n_{nl} - n_{fl}}{n_{fl}} \times 100\%$$

- Most prime movers have a speed drop from 2% to 4%. Most governors have a mechanism to adjust the turbine's no-load speed.



- The relationship between frequency and power can be described:

$$P = s_p (f_{nl} - f_{sys})$$

P : Power output of the generator

f_{nl} : No-load frequency of the generator

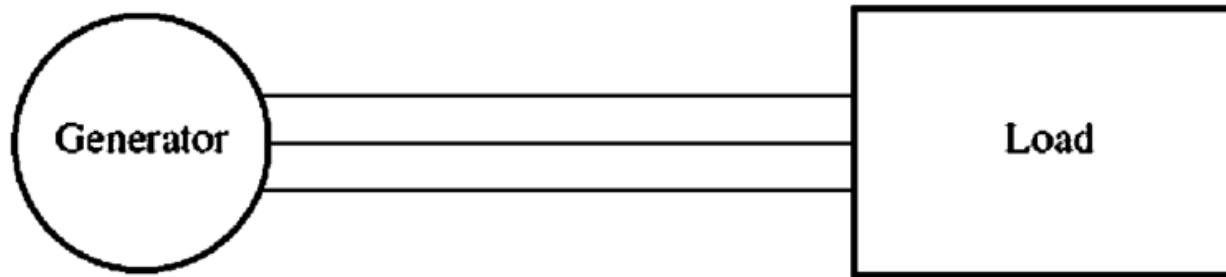
f_{sys} : Operating frequency of system

s_p : Slope of curve, in kW/Hz or MW/Hz

12. Generator Operating Alone

When a generator is operating alone supplying the load:

- The real and reactive powers are the amounts demanded by the load.
- The governor of the generator controls the operating frequency of the system.
- The field current controls the terminal voltage of the power system.

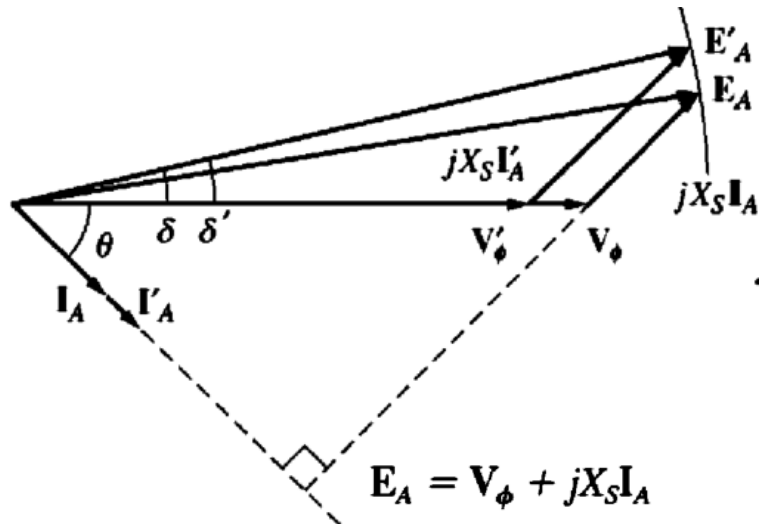


What happens when we increase the load on this generator?

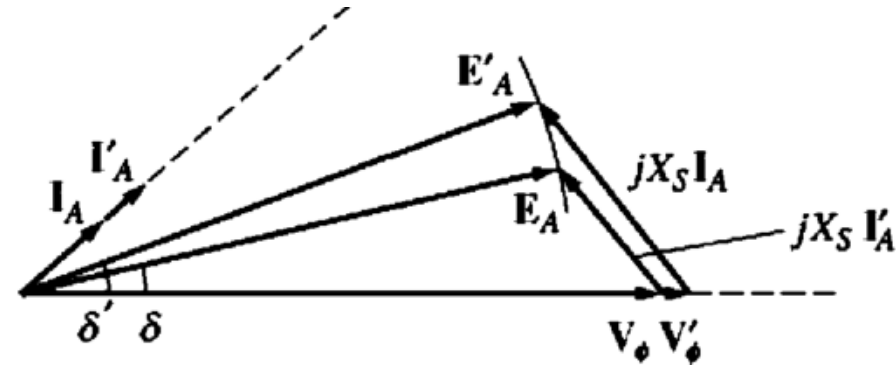
- An increase in the load is an increase in the real and/or reactive power drawn from the generator.
- Such a load increases the load current drawn from the generator.
- Because the field resistor has not been changed, the field current is constant, and therefore the flux is constant. Since the prime mover also keeps a constant speed.
- The magnitude of the internal generated voltage $E_A = K\phi\omega$ is constant.

12. Generator Operating Alone

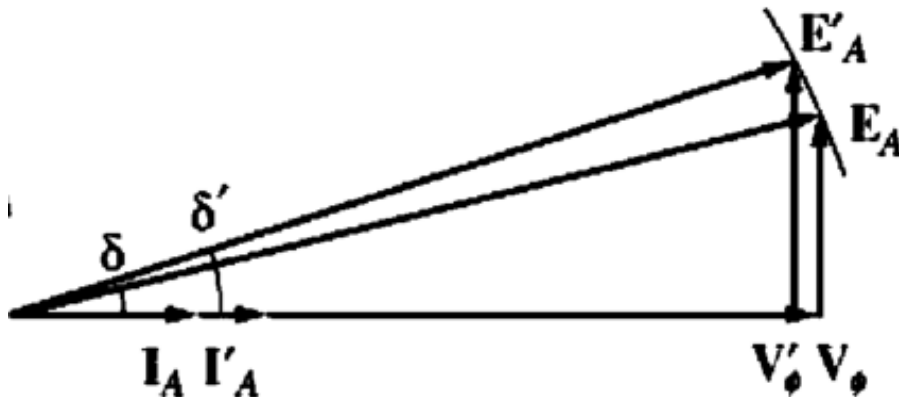
- A generator operating at a lagging power factor:



- Let the generator be loaded with leading-power-factor loads:



- Now suppose the generator is loaded with unity-power-factor loads:



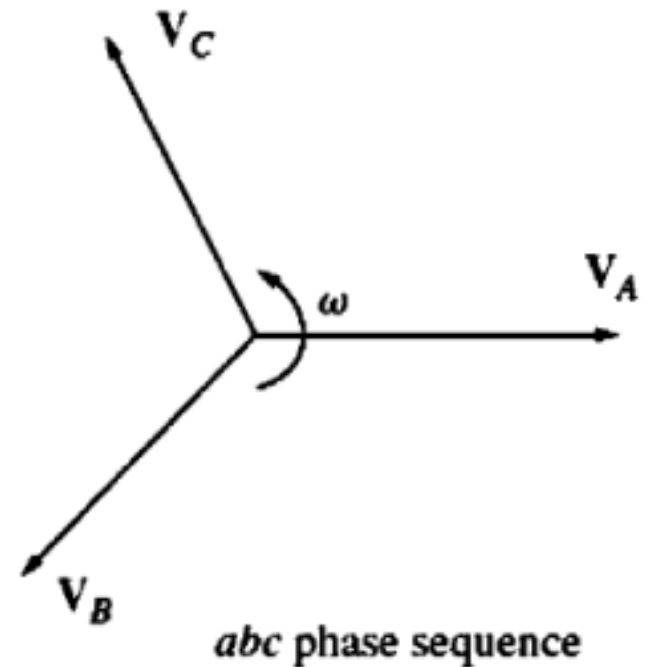
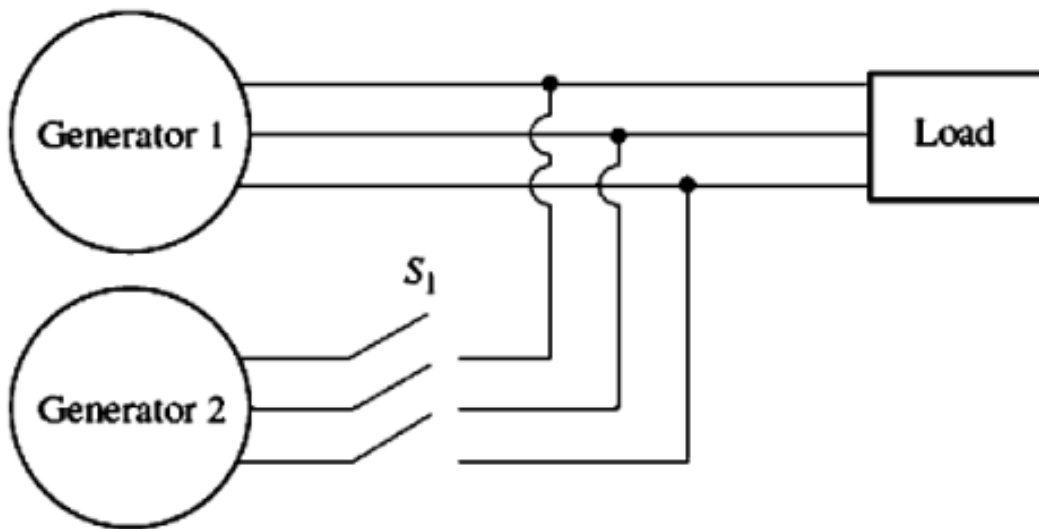
13. Parallel Operation of AC Generator

- In today's world, an isolated synchronous generator supplying its own load independently of other generators is very rare. For all usual generator applications, there is more than one generator operating in parallel to supply the power demanded by the loads.
- There are several major advantages to such operation:
 - Several generators can supply a bigger load than one machine by itself.
 - Having many generators increases the reliability of the power system, since the failure of anyone of them does not cause a total power loss to the load.
 - Having many generators operating in parallel allows one or more of them to be removed for shutdown and preventive maintenance.
 - With many generators operating in parallel, depending on the load condition, allows the operation near full load then maximum efficiency can be obtained.

13. Parallel Operation of AC Generator

The Conditions Required for Paralleling:

- The RMS line voltages of the two generators must be equal.
- The two generators must have the same phase sequence.
- The phase angles of the two phases must be equal.



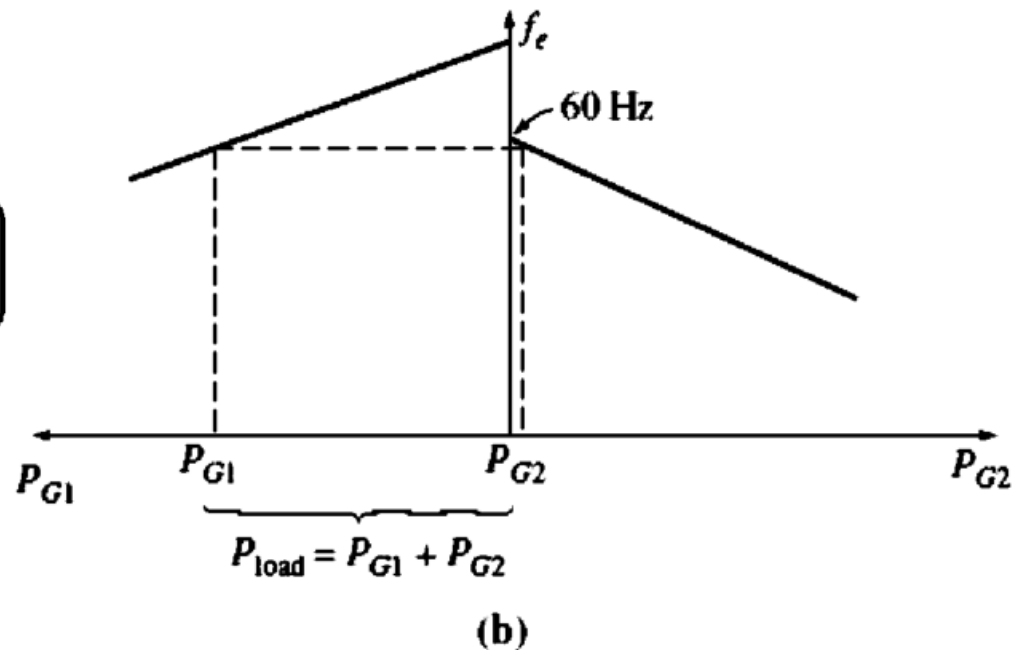
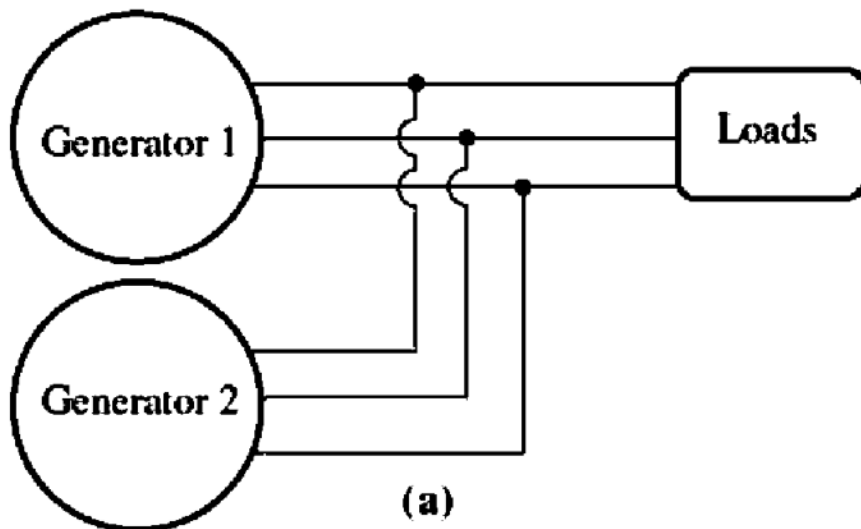
13. Parallel Operation of AC Generator

Parallel operation of two Generator:

- The power-frequency diagram right after G2 is connected to the system is shown to the right.
- Note that the sum of the real and reactive powers supplied by the two generators must equal the real and reactive powers demanded by the load.

$$P_{\text{tot}} = P_{\text{load}} = P_{G1} + P_{G2}$$

$$Q_{\text{tot}} = Q_{\text{load}} = Q_{G1} + Q_{G2}$$

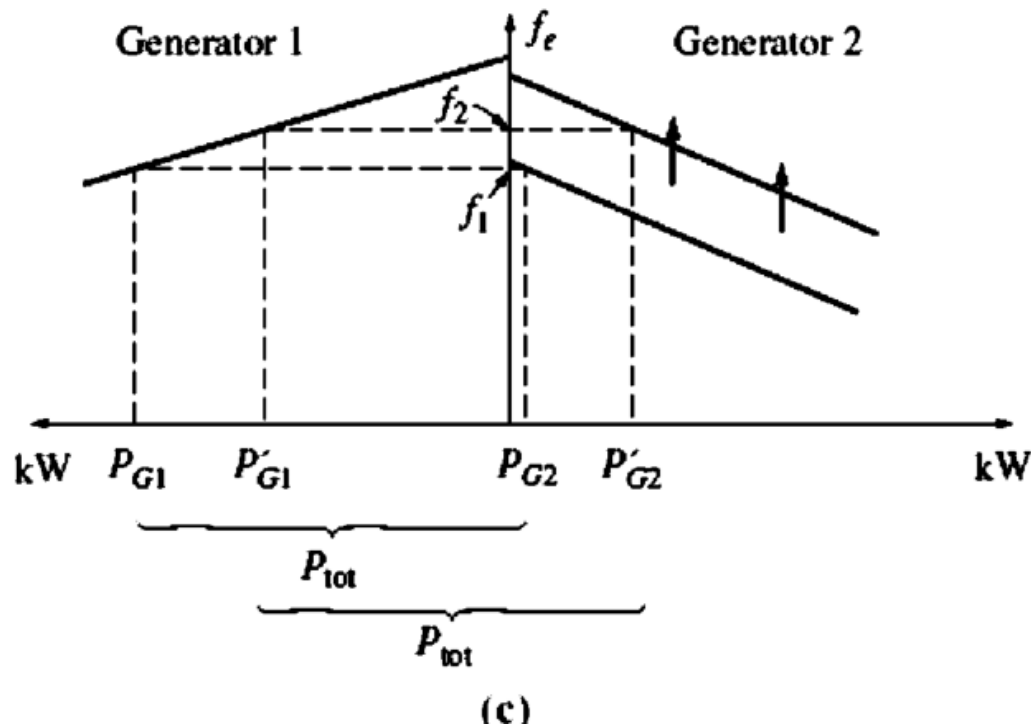


13. Parallel Operation of AC Generator

Parallel operation of two Generator:

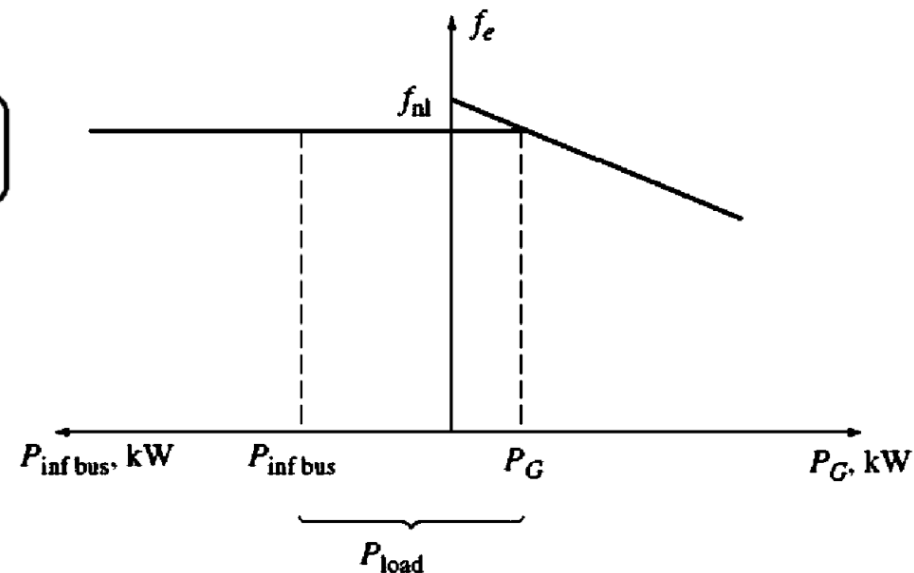
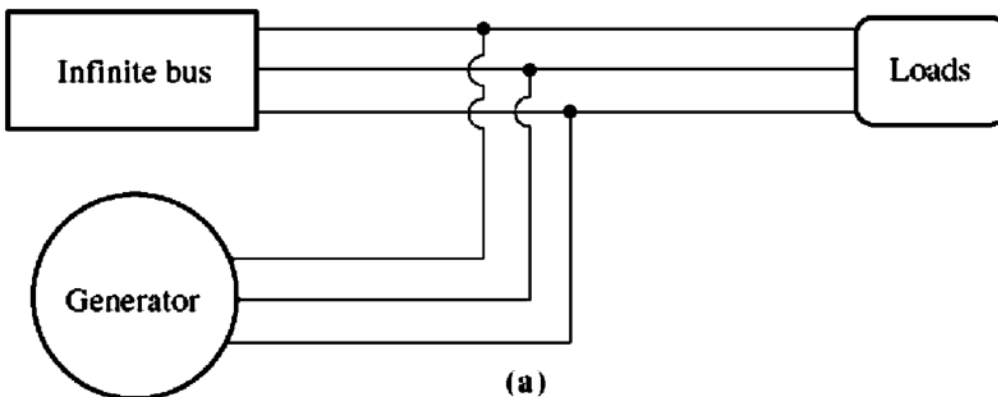
If the speed of G2 is increased, its power-frequency diagram shifts upwards. This will in turn:

- Increase the real power supplied by G2
- Reduce the real power supplied by G1
- Increase the system frequency



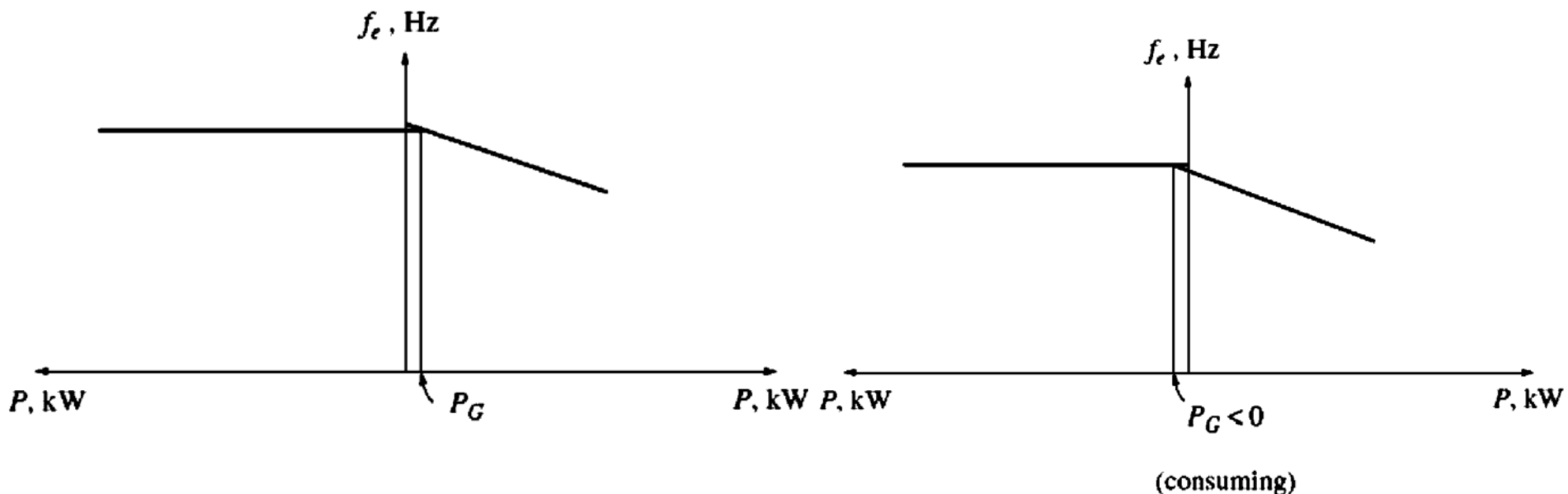
14. Synchronizing a Generator with the Utility grid

- When a synchronous generator is added to a power system, that system is so large that one additional generator does not cause observable changes to the system.
- Consider adding a generator to an infinite bus supplying a load.
- The frequency and terminal voltage of all machines must be the same. Therefore, their power-frequency and reactive power-voltage characteristics can be plotted with a common vertical axis.



14. Synchronizing a Generator with the Utility grid

- If the no-load frequency of the oncoming generator is slightly higher than the system's frequency, the generator will be floating on the line supplying a small amount of real power and little or no reactive power.
- If the no-load frequency of the oncoming generator is slightly lower than the system's frequency, the generator will supply a negative power to the system. The generator actually consumes energy acting as a motor. Many generators have circuitry automatically disconnecting them from the line when they start consuming energy.



Example 4:

Figure (a) below shows two generators supplying a load. Generator 1 has a no-load frequency of 61.5 Hz and a slope s_{p1} of 1 MW/Hz. Generator 2 has a no-load frequency of 61.0 Hz and a slope s_{p2} of 1 MW/Hz. The two generators are supplying a real load totaling 2.5 MW at 0.8 PF lagging. The resulting system power-frequency or house diagram is shown in Figure (b).

1. At what frequency is this system operating, and how much power is supplied by each of the two generators?
2. Suppose an additional 1-MW load were attached to this power system. What would the new system frequency be, and how much power would G_1 and G_2 supply now?
3. With the system in the configuration described in part b, what will the system frequency and generator powers be if the governor set points on G_2 are increased by 0.5 Hz?

